

A Blockchain Sharding-Based Data Sharing Scheme for Internet of Vehicles

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Abstract—In recent years, the adoption of blockchain technology has significantly advanced distributed trust and privacy protection, rendering it a popular choice for data sharing scenarios within the Internet of Vehicles (IoV). However, when applied to data sharing contexts, the technology encounters issues related to low throughput and sluggish consensus speed. To address these issues, this paper proposes a novel data sharing scheme tailored for IoV that leverages a sharded blockchain to boost system throughput. The scheme dynamically selects nodes and shards based on vehicle reputation, computing power, and location centrality, ensuring the security of shard verification. Additionally, the scheme incorporates a proxy node selection strategy and utilizes Boneh-Lynn-Shacham (BLS) signature aggregation technology to optimize the Practical Byzantine Fault Tolerance (PBFT) consensus mechanism. Experimental results demonstrate that, compared with other schemes, the proposed approach achieves superior levels of security, reliability while maintaining lower communication complexity.

Index Terms—Internet of vehicles, data sharing, blockchain, dynamic sharding, consensus mechanism

I. INTRODUCTION

With the rapid advancement of the Internet of Vehicles (IoV), data security concerns have become increasingly significant [1]. The risks associated with false data and privacy breaches pose significant threats to the security and efficiency of data sharing. While data encryption/decryption techniques (e.g., via chaotic secure communication [2],[3], data coding [4],[5], etc.) can ensure the secure transmission and storage of information in vehicular networks, their algorithmic implementations are complex and rely on high-quality channels with limited anti-interference capabilities. In this situation, there is a need for a method that can both protect data security and cope with the complexity of the algorithms and the limitations of channel quality. This underscores the importance of ensuring the privacy, authenticity, and integrity of IoV data.

In recent years, blockchain has emerged as an effective solution to data privacy and trust issues due to its decentralized, distributed, anonymous, and tamper-resistant characteristics, making it a popular choice for data sharing scenarios in IoV [6]. Most of the existing literature deploys a single layer of blockchain at the RSUs level. While this design strategy simplifies the system architecture, it also introduces some issues [7], [8], [9]. Firstly, the system throughput decreases as the number of network nodes increases. This is because all transactions in the blockchain system need to be verified and stored on all nodes, and as the number of nodes increases, system performance and efficiency decrease. Secondly, existing solutions do not restrict the joining and exiting of nodes, which provides the possibility for malicious nodes to launch attacks. In P2P networks, nodes can freely

join and leave, which provides a possibility for malicious nodes, they may launch collusion attacks or eclipse attacks, affecting the normal operation of the blockchain [10].

To address these challenges, this paper proposes a secure and reliable data sharing scheme built upon sharded blockchain technology. The adoption of blockchain sharding is driven by the necessity for high transaction throughput, a critical requirement for handling real-time, large-scale data generated and utilized within vehicular networks. By subdividing the network into smaller pieces, known as “shards”, each capable of independently processing its own transactions and smart contracts, we significantly enhance the scalability of our solution.

In addition to scalability improvement, maintaining robustness of the consensus mechanism, especially against Sybil attacks, is equally significant. Traditional Byzantine Fault Tolerance (BFT) systems, while robust, have high communication overheads and are susceptible to Sybil attacks. PBFT is a consensus algorithm ensuring agreement among nodes in distributed systems, even with up to $(n - 1)/3$ faulty or malicious nodes. In view of this, we introduce an optimized BLS-PBFT consensus algorithm. BLS signatures [11] are short and easy to verify, reducing validation time and communication overhead, while enhancing resistance to Sybil attacks. This By combining the robustness of PBFT and the efficiency of BLS signature aggregation, the overall security and efficiency of our solution can be improved.

The main contributions of this work can be summarized as follows:

- We propose a sharded blockchain-based data sharing scheme, enhancing data security and reliability.
- We propose a blockchain sharding protocol and a proxy node selection algorithm, reducing risks associated with malicious nodes and improving overall system security.
- We introduce a BLS-PBFT consensus algorithm for IoV scenarios that improves transaction verification efficiency, reduces communication overhead, and increases Sybil attack.

II. RELATED WORKS

In this section, we review the state of the art in two key areas that underpin our research: the application of blockchain technology in IoV data sharing and the use of the PBFT consensus algorithm.

A. Blockchain in IoV Data Sharing

Blockchain technology in IoV data sharing has gained significant attention. For example, Cui *et al.* [7] designed an enhanced DPoS algorithm, reducing communication delay for

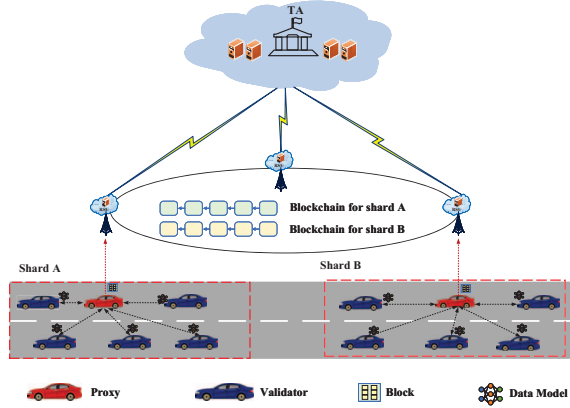


Fig. 1. System Model

distributed IoV applications. Wang *et al.* [8] proposed a multi-sharding blockchain framework for vehicular data privacy, improving system throughput. Kang *et al.* [12] proposed a data sharing scheme combining blockchain and a three-weight subjective logic reputation model for secure, efficient data sharing among vehicles. However, their methods did not account for the significant decrease in data transaction throughput as the number of nodes increases. Furthermore, their solutions did not fully consider potential collusion and eclipse attacks. In comparison, our approach, based on sharded blockchain, offers a more effective response to these challenges, segmenting the network to enhance the scalability of the solution.

B. Improved PBFT Consensus for IoV

In the context of the Internet of Vehicles (IoV), consensus algorithms play a vital role in upholding the integrity and reliability of traffic-related data. The PBFT consensus algorithm stands out prominently due to its robustness against Byzantine faults. For example, Diallo *et al.* [13] improved the PBFT consensus protocol for vehicular networks for better security and efficiency. Xu *et al.* [14] proposed a SG-PBFT consensus protocol for the IoV, improving the traditional PBFT algorithm by using a score grouping mechanism for better consensus efficiency. Kumar *et al.* [15] proposed an HN-mPBFT protocol for the IoV, streamlining PBFT consensus phases to minimize message overhead and transaction latency. The complexity of these improved PBFT algorithms are $O(n^2)$, limiting their suitability for high-throughput, fast-paced IoV scenarios. In this work, we propose a lightweight BLS-PBFT approach which reduces its complexity to $O(n)$, better accommodating high-speed IoV environments.

III. SYSTEM MODEL

A. System Model

In our considered IoV data sharing network (as illustrated in Fig. 1), there are three main entities: trusted authority (TA), roadside units (RSUs), and vehicles. To ensure system security, RSUs communicate with the TA and with each other via wired transmission. Additionally, all RSUs and vehicles must register with the TA prior to joining the network. The roles and functions of these three entities are described in details as follows.

Trusted Authority (TA): The TA serves as an authoritative third party responsible for verifying and authorizing the identity and conduct of vehicles and RSUs. Its primary role is to ensure the security and integrity of data transmission within the network. Typically, the TA is an entity designated by the government or other organizations, entrusted with the core task of upholding the security and trustworthiness of the vehicular network.

RSUs: The RSU is an infrastructure device deployed alongside roadways, designed for collecting and relaying information sent by vehicles and TA. It also actively participates in global blockchain consensus process and maintain storage of all shard blockchain data.

Vehicles: Vehicles in the data sharing system have computational resources, storage, and wireless communication, enabling them to collect, process, validate, and transmit data. However, their trustworthiness may vary as they could potentially forge identities or provide false data for rewards.

B. Blockchain Model

In our scheme, vehicles initially perform local data processing before employing the blockchain for intra-shard consensus on the data model. Data models that meet the required criteria are then aggregated into a global model for upload, which is subsequently uploaded to Roadside Units (RSUs) for comprehensive validation. The blockchain consensus model encompasses four distinct roles: master node, global node, proxy nodes, and validator nodes. Specifically, the master node corresponds to the TA, global nodes corresponds to RSUs, while the proxy and validator nodes are comprised of the vehicles themselves.

1) Master node: Tasked with inspecting and managing node statuses, the master node allocates roles to network nodes based on sharding and proxy selection algorithms, assigning them to their respective consensus shards.

2) Global node: Global nodes, primarily composed of Roadside Units (RSUs), are responsible for validating shard blockchain data. They also synchronize and store all the data from these shard blockchains.

3) Proxy node: Proxy nodes collect feedback results from shard verification nodes, generate the intra-shard block, and disseminate it to the global nodes.

4) Validator node: Validator nodes receive unverified data blocks from proxy nodes and verify their compliance with the data collection criteria. They also monitor the activities of proxy nodes.

IV. THE PROPOSED SCHEME

In this section, we present our proposed system scheme and describe how vehicles share data and verification messages in our blockchain system. Specifically, the workflow of the data sharing scheme base on BFL is described as follows.

1) Setup: During system initialization, each vehicle and RSU generate a unique public-private key pair (pk_i^v, sk_i^v) , which is then submitted to the TA for registration. Upon registration, each node receives a unique ID number and a digital certificate Cv_i . Before joining the network, vehicles must send this information to the RSU for identity verification, ensuring secure and authenticated communication within the network.

$$ID_i^v = \text{hash}(pk_i^v), \quad (1)$$

DataSet	SigSet	DataDecSet	DataDecMap
DataHash	DataSetHash	DataHash	DataSetHash
hash(data ₁) sig _{d₁}	hash(DataSet ₁) sig _{n₁}	Data ₁ Dec	DataSet ₁ Dec sig _{n₁}
hash(data ₂) sig _{d₂}	hash(DataSet ₂) sig _{n₂}	Data ₂ Dec	DataSet ₂ Dec sig _{n₂}
...	...	...	...
hash(data _p) sig _{d_p}	hash(DataSet _p) sig _{n_p}	Data _p Dec	DataSet _p Dec sig _{n_p}
CS _i	CS _i	CS _i	CS _i
Sig _i	Sig _i	Sig _i	Sig _i
Timestamp	Timestamp	Timestamp	Timestamp

Fig. 2. Data structure of DataSet, SigSet, DataDecSet and DataDecMap

2) *Proxy Selection*: In the data sharing blockchain system, nodes are categorized as proxy and validator nodes based on their consensus scores (CS). At the beginning of data sharing, the master node instructs each node to provide its CS , calculated using smart contracts. Nodes failing to do so are labeled as selfish by the master node and excluded from consensus processes.

The CS considering reputation score (RS), computing power (CP), and location centrality (LC) of each node. It then sorts these scores to obtain CS_{sort} . The deep reinforcement learning method, as outlined in [16], is used to determine both the number of shards k and the proxy threshold CS_{th} .

The master node selects k nodes with CS scores exceeding CS_{th} to serve as proxy nodes, based on geographical locations, and issues authorization certificates. Upon receiving the certificates, proxy nodes can transmit beacons to other nodes within range, declaring their identity. Proxy nodes are required to record their CS scores in the block during the consensus process for facilitating future view changes throughout the data sharing period. Details are presented in Fig. 2.

$$CS = \alpha RS + \beta CP + (1 - \alpha - \beta) LC, \quad (2)$$

where α represents the weight factor for the RS , and β represents the weight factor for the CP .

a) RS : The RS depends on the behavior of the node and the historical reputation score. This implementation uses a Beta distribution to calculate the RS , with the following specific steps are as follows:

Step 1. Initialization: At the start, an initial Beta distribution is assigned to each node $node_i$. The number of honest behaviors pos_i is set to 1, the number of malicious behaviors neg_i is set to 1, and the historical behavior list H_i is set to be an empty set $[\emptyset]$ initially. For example, $Beta(pos_i = 1, neg_i = 1)$, in this implementation, the initial preset for node behavior is a malicious probability of 0.5.

Step 2. Node behavior observation: At each time t , we observe the behavior of node i , $b_{i,t}$. If the behavior of the node is confirmed to be honest, $pos_{i,t} = pos_{i,t-1} + 1$. If the behavior of the node is confirmed to be malicious, then $neg_{i,t} = neg_{i,t-1} + 1$. Here, $pos_{i,t-1}$ and $neg_{i,t-1}$ represent the number of honest and malicious behaviors at the previous time $t - 1$, respectively. We add the behavior $b_{i,t}$ to the end of the historical behavior list H_i .

Step 3. Introduction of a decay factor: For each node i , after updating pos and neg at each time t , this implementation multiplies them both by a decay factor γ (for example, $\gamma = 0.9$). This adjustment ensures that earlier behaviors carry less weight, while recent behaviors have a larger weight indicating greater influence.

$$pos_{i,t} = \gamma * pos_{i,t}, \quad (3)$$

$$neg_{i,t} = \gamma * neg_{i,t}, \quad (4)$$

Step 4. Sliding window update: This implementation sets a sliding window N , for example, with $N = 10$. The implementation takes into account only the most recent N behaviors to update pos and neg . If the length of the historical behavior list H_i exceeds N , the implementation removes the first element $b_{i,t-N}$ from the list H_i and update pos and neg accordingly based on its value. If $b_{i,t-N}$ is honest, then $pos_{i,t} = pos_{i,t-1}$. If $b_{i,t-N}$ is malicious, then $neg_{i,t} = neg_{i,t-1}$.

Step 5. Reputation score calculation: At each time t , this implementation can calculate the reputation score RS_i of node i as

$$RS_i = \frac{pos_{i,t}}{pos_{i,t} + neg_{i,t}}, \quad (5)$$

Step 6. Reputation score update: Repeat steps 2 to 5, continuously observe the behavior of nodes, and update their reputation scores accordingly.

b) CP : The CP represents the specific computational capability of a node used for data verification in consensus, reflecting the number of instructions that the node's CPU can execute per unit of time. Therefore, CP can be given as

$$CP = H(s, \tau), \quad (6)$$

where s represents the number of instructions and τ represents the time required to execute the task. $H(s, \tau)$ represents a function with respect to s and τ .

c) LC : The LC represents the positional centrality of nodes in a network, which depends on the degree centrality and closeness centrality of the nodes. Therefore, LC of node v_i can be given as

$$LC_i = \eta \frac{n-1}{\sum_{j=1}^n a_{ij}} + (1-\eta) \frac{n-1}{\sum_{j=1}^n d_{ij}} \\ = (n-1) \left[\frac{\eta}{\sum_{j=1}^n a_{ij}} + \frac{1-\eta}{\sum_{j=1}^n d_{ij}} \right], \quad (7)$$

where n represents the total number of nodes in the network, a_{ij} represents the elements of the adjacency matrix A . When there is an edge between node i and node j , $a_{ij} = 1$. Otherwise $a_{ij} = 0$. d_{ij} represents the shortest path length between node i and node j . η is a weight parameter used to configure the importance of degree centrality and closeness centrality in the calculation of locational centrality.

3) *Formation of Shards*: Within one epoch t , each proxy node maintains a single shard. To mitigate security risks associated with fewer nodes in a shard, every validator node can choose to join up to s ($s < k$) shards. Upon receiving beacons from proxy nodes, a validator node evaluates their reputation, positional distance, and shard size indicators F_S . It then generates a candidate list for shard participation based on this assessment, subsequently joining various shards in accordance with this list.

Proxy nodes persistently monitor the average CS score and its trend within the shard. The CS score is a reputation system that measures node behavior, where nodes with higher scores are more likely to be honest, while those with lower scores could be malicious or poorly performing nodes. If the score diminishes rapidly, this could imply that malicious activities are taking place within the shard or some nodes are underperforming. To safeguard the system's security and performance, the proxy node rejects applications from validator nodes with lower CS scores when a rapid drop in score is detected. This action prevents potentially malicious nodes from joining the shard and initiating attacks.

$$FS = \lambda_1 RS + \lambda_2 DS + (1 - \lambda_1 - \lambda_2)SZ, \quad (8)$$

Here, RS denotes the proxy node's reputation score, DS is the positional distance between the proxy node and the validator node, and SZ signifies the current shard size (maximum number of nodes participating in consensus within the shard). The shard size under a proxy node's purview expands with its rising CS score. λ_1 and λ_2 respectively represent the weight factors of the reputation score RS and distance score DS , respectively.

4) *Task Issuance by TA*: Based on the operational needs of Intelligent Transportation Systems, TA issues data collection tasks. The aim of these tasks may cover multiple aspects, such as collecting traffic flow data, road condition information, traffic accident reports, etc. These data are of significant value for improving road safety, reducing traffic congestion, and more.

5) *Vehicle Training Data*: Upon receiving the task, vehicles start data collection and training. After signing the pre-trained data model, vehicles send it to the proxy node.

6) *Block Generation to be Verified*: Upon signing the pre-trained data model, the vehicle shares it with the proxy node. The proxy node gathers the data and creates a micro-block, prepared for verification.

7) *Intra-shard Consensus Verification of Data Models Based on BLS-PBFT*: In our proposed scheme, we leverage the BLS-PBFT consensus mechanism to verify the data model in each shard. The traditional PBFT consensus mechanism has four stages: pre-prepare, prepare, commit and reply. In order to reduce the communication complexity, we introduce BLS signature technology to significantly reduce the communication delay during the prepare and commit stages. Moreover, to discourage malicious behavior of the shard proxy, we design a reporting mechanism after the reply stage. The communication pattern of our proposed BLS-PBFT consensus is demonstrated in Fig. 3 and BLS-PBFT consensus steps are presented as follows.

Step 1. The proxy node receives packets from the vehicle and packages them into a data block referred to as *DataSet*.

Step 2. The proxy node distributes the *DataSet* to all validators.

Step 3. Each validator signs the received data and sends it back to the proxy node.

Step 4. The proxy node aggregates signatures using BLS and organizes them into a signed data block referred to as *SigSet*, which is then broadcast to all validators.

Step 5. Each validator examines the correctness of all signatures in the *SigSet*, verifies whether the precision of each data model meets the requirements, and assesses the

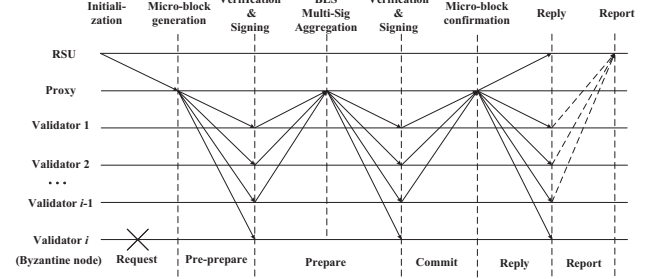


Fig. 3. Communication pattern of the BLS-PBFT consensus

timeliness of the data. Subsequently, they reach a decision (i.e., either acceptance or rejection), compile this decision into a data validation result block referred to as *DataDecSet*, sign it, and then return this *DataDecSet* to the proxy node.

Step 6. The proxy node packages all the *DataDecSets* into a data validation result collection block called *DataDecMap*, and then sends the *DataDecMap* to the nearest RSU and all the validators.

Step 7. Upon receipt of the *DataDecMap*, the RSU verifies whether the number of *DataDecSets* meets the valid consensus threshold κ , and it approves the block if this condition is met. The value of κ is determined based on factors including the node count within the shard n , the average CS score, and the variance of the CS scores. A larger node count corresponds to a larger κ value. If the average CS score is low and the variance of the CS scores is small, this may be indicating a potential collusion attack, κ is increased accordingly. The specific calculation of κ is given in Eq. (9).

$$\kappa = \theta * n * (1 - \chi * \frac{\sum_{i=1}^n CS_i}{n} * \phi * \frac{\sum_{i=1}^n (CS_i - \frac{\sum_{i=1}^n CS_i}{n})^2}{n-1}), \quad (9)$$

where CS_i represents the individual CS score of node i within the shard. θ represents the influence factor for the number of nodes, χ represents the influence factor for the average CS score of all nodes, and ϕ represents the influence factor for the variance of CS scores.

The RSU synchronizes this value with all vehicles in the shard and appends the block to the local blockchain. Meanwhile, the validator node thoroughly inspects the data for any anomalies and alerts the master node if any issues are detected.

8) *Global Consensus and Data Synchronization*: After consensus has been reached within each shard, a global consensus validation process is carried out among the global nodes. During this stage, the consensus-reached shard blocks are broadcasted to all global nodes. Subsequently, the global nodes verify whether the number of *DataDecSets* has reached the effective consensus threshold κ . If the shard block is deemed valid, the global node sends the valid result to other global nodes. If other global nodes receive feedback indicating that more than 2/3 of the results align with their own, they proceed to include the shard in the global blockchain and store all the associated data.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Simulation area size	4km × 4km
Communication protocol	LTE-V2X
Speed of vehicles	9 ~ 90 km/h
Number of vehicles, N	20,30,40,50,60,70,80
CS score threshold, CS^{th}	0.5
Number of shards, k	1 ~ 20
Maximum number of selectable shards, s	1 ~ 10
CS score weight parameter	$\alpha = 0.4, \beta = 0.3$
FS score weight parameter	$\lambda_1 = 0.4, \lambda_2 = 0.4$

V. SIMULATION AND ANALYSIS

In order to evaluate the efficacy of the proposed scheme, we conduct simulation tests on the Veins platform. We leveraged OMNeT++ and SUMO as our simulation tools and adhered to the setup parameters listed in Table I to conduct our simulations.

In our experiments, we focused on analyzing three key performance indicators, namely security, delay and throughput. We performed a total of 100 simulation runs to ensure the reliability and statistical significance of our evaluation results, and the reported results are averaged obtained from these runs.

A. Security

Our security metric gauges the impact of varying blockchain validation schemes on data sharing security under varying proportions of malicious vehicles. We conducted simulations for over 10,000 data sharing scenarios to observe the effectiveness of different schemes in preventing malicious data sharing. As Fig. 4 depicts, our scheme nearly attains

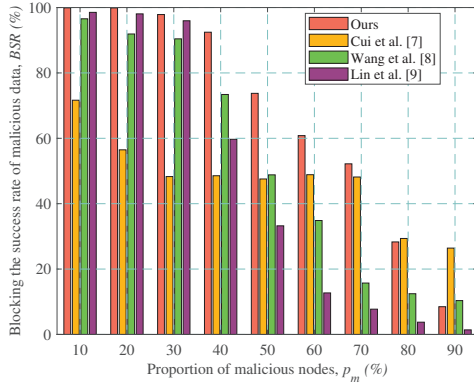


Fig. 4. success rates in blocking malicious data among ours and other blockchain-based data sharing schemes.

100% success with 10%-30% malicious nodes, and it continues to outperform other approaches even when faced with 70% malicious vehicles. This success can be attributed to our scheme's multi-shard consensus verification by honest vehicles, which effectively reduces the malicious vehicle ratio per shard.

However, it's worth noting that the DPoS consensus algorithm in [7] may be vulnerable to vote bribery, which can impact its security. While both our scheme and the scheme proposed in [8] demonstrate more robust resistance to a

growing number of malicious nodes better, all schemes falter when malicious nodes exceed 80%.

B. Consensus delay

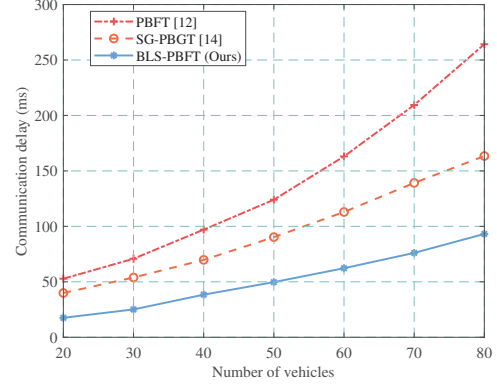


Fig. 5. Communication delay comparison among PBFT consensus [12], SG-PBFT consensus [14] and BLS-PBFT consensus.

The consensus delay refers to the average time required for data consensus verification, which was tested for traditional PBFT [12], SG-PBFT [14], and the BLS-PBFT proposed in this paper, and conducted a comparative analysis. Fig. 5 shows the communication delay of the three different blockchain consensus algorithms under varying numbers of vehicles. The experiments were repeated 1000 times, and the average time delay was recorded to improve the credibility of the result data. It can be seen from the figure that the BLS-PBFT consensus proposed in this paper outperforms traditional PBFT and SG-PBFT consensus significantly, especially as the number of vehicles increases. This improvement can be mainly attributed to the utilization of BLS signature aggregation technology in the prepare and commit phases.

C. Throughput

We evaluated the sharding scheme's throughput, comparing the rates without sharding [9] and with multi-sharding [8]. Vehicles sent 500 data models per second. After 100 blocks, we compared the average transactions per second (TPS) under different numbers of vehicles.

As shown in Fig. 6, our results demonstrated that our scheme outachieves both non-sharding and multi-sharding schemes in terms of TPS. This superiority is attributed to the use of the BLS-PBFT consensus algorithm in our scheme, which streamlines the consensus process, and the use of CS sharding significantly enhances the system's average consensus verification efficiency. While the multi-sharding scheme does boost system throughput, the additional latency brought about by its multi-round inter-shard communications offsets some of its advantages.

As shown in Fig. 7, our proposed BLS-PBFT consensus algorithm outperforms the SG-PBFT [14] and HN-mPBFT [15] algorithms in high-speed scenarios. This is because that the use of BLS aggregated signature algorithm can reduce communication complexity and significantly enhance system throughput.

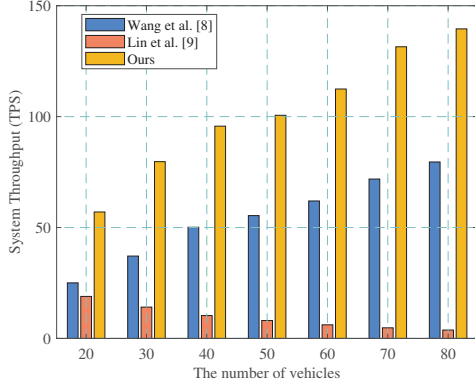


Fig. 6. Throughput comparison among the proposed schemes in [8], [9].

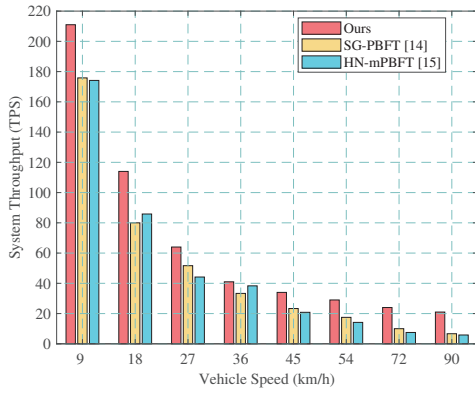


Fig. 7. Comparison of consensus algorithm throughput under varying vehicle speeds.

VI. CONCLUSION

This paper proposes a blockchain sharding algorithm aimed at enhancing data sharing efficiency upholding upholding data security within the dynamic IoV environment. The algorithm achieves this by segmenting consensus resources and transforming the traditional single consensus unit into multiple decentralized consensus units, leveraging factors such as reputation and computing power. Specific roles are allocated to nodes based on a proxy node selection strategy, resulting in increased reliability and reduced view switching. Additionally, Boneh-Lynn-Shacham (BLS) signature technology is incorporated to address communication challenges within alliance chain. Experimental results demonstrate superior security, reliability, and reduced communication complexity compared to alternative data sharing schemes.

In the future, we will explore integrating this sharding mechanism with edge computing to further enhance its scalability in large-scale IoV.

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